Build Your Own Data Analysis Pipeline - Capstone Project

Student Information

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Project Overview

This project focuses on analysing the Ames Housing dataset to understand the main factors that affect house prices. The work follows a structured pipeline including data cleaning, feature engineering, and exploratory data analysis.

Data Cleaning

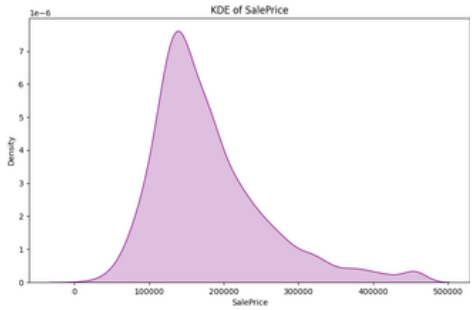
- Loaded the dataset using pandas
- Explored the data using .info(), .head(), and .shape()
- Converted data types for relevant columns
- Handled missing values by dropping high-missing columns and filling others with median
- Removed duplicate rows
- Treated outliers in SalePrice using the 99th percentile
- Added validation checks to ensure data quality

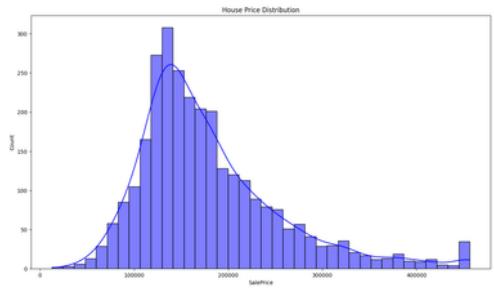
Feature Engineering

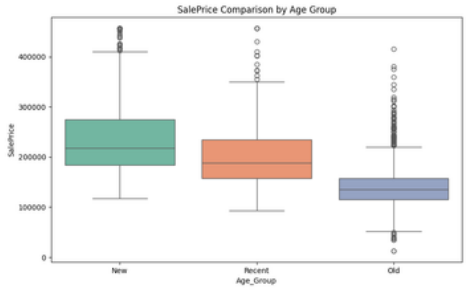
- Applied one-hot encoding on categorical columns
- Converted Exter_Qual into numerical values
- Scaled numerical features using StandardScaler
- Created new features:
  - `Pri ce_ Pe r_S qft`
  - `House_Age`
  - `Qua lit y_ x_A rea`
- Applied log transformation on SalePrice
- Binned house age into groups (New, Recent, Old)
- Removed highly correlated features

Exploratory Data Analysis (EDA)

- Visualised the distribution of house prices using histograms and KDE
- Analysed the relationship between living area and price using scatter plots
- Identified top correlated features with price
- Compared price distributions across different house age groups
- Analysed average prices across neighborhoods







Math & Analysis

- Calculated mean and standard deviation manually
- Applied standardization using mathematical formula
- Estimated probabilities based on conditions
- Used cosine similarity to compare data points

Conclusion

The analysis shows that house size, quality, and location are key factors influencing house prices. Feature engineering and transformations improved the understanding of the data and provided clearer insights.